

MAPS Energy and Materials Module (Deliverable D4.2)

****Eff-TFP Analysis Kit (R) + Integrated Energy-Economy Model (IEEM, Excel)****

MAPS Project – Horizon Europe Grant Agreement No. 101137914

This repository contains two interconnected tools that together form the ****Energy and Materials Module**** of the MAPS project:

1. ****Eff-TFP Analysis Kit**** – an R script that empirically tests the long-run relationship between aggregate final-to-useful exergy efficiency (ϵ) and Total Factor Productivity (TFP), and estimates the useful exergy intensity of GDP.
2. ****Integrated Energy-Economy Model (IEEM)**** – an Excel-based dynamic simulation tool that uses the Eff-TFP relationships to project GDP, energy demand, renewable capacity, and CO₂ emissions under Business-as-Usual and Decarbonisation scenarios.

The two components are designed to be used sequentially:

- ****Step 1**** – Run the Eff-TFP kit on historical data to obtain the country-specific parameters:
 - Cointegration coefficient (β) linking TFP to exergy efficiency: $\ln(\text{TFP}) = \beta \cdot \ln(\epsilon)$ (Hz specification, quality-adjusted TFP).
 - Useful exergy intensity: $X_{\text{useful}} / \text{GDP}$ (tested for constancy).
- ****Step 2**** – Input these parameters into the IEEM Excel tool, which then simulates future GDP, energy use, and emissions under alternative scenarios.

1. Eff-TFP Analysis Kit (R script)

What it does

- Loads macroeconomic, capital, labour, and exergy data (or pre-computed TFP/efficiency series).
- Computes output elasticities (α_K , α_L) from income shares (or uses defaults 0.3/0.7).
- Constructs unadjusted and quality-adjusted TFP via a Cobb-Douglas production function.

- Tests unit roots (ADF & PP) and runs Johansen cointegration (Hz and Hc specifications).
- Estimates the long-run elasticity of TFP with respect to exergy efficiency.
- Tests constancy of useful exergy intensity.
- Outputs a multi-sheet Excel workbook and a formatted PDF report.

Data input – two templates

The kit auto-detects which template you are using.

Standard Template (`1\_Standard\_Template`)

Provide full macroeconomic and exergy time series. The kit computes TFP from scratch.

| Required columns | Optional (improves quality) |

|-----|-----|

| `year`, `gdp`, `k\_stock`, `emp`, `x\_final`, `x\_useful` | `k\_services`, `avh`, `hc`,
`adj\_gos`, `gos`, `coe`, `wap`, `lfpr`, `ur` |

> ***“Standard Mode — provide all macro + exergy columns. The kit computes TFP from scratch using factor shares, labour, capital, and GDP. Yellow = optional (used if available; analysis proceeds without them).”** (from Excel template)

Pre-loaded Template (`2\_Preloaded\_Template`)

Use this if TFP and efficiency series are already available (e.g. from a previous run or external source). The kit skips steps 2–5 and only performs cointegration, diagnostics, and reporting.

| Required columns |

|-----|

| `year`, `tfp\_unadj` (or `z\_tfp\_unadj`), `eff` (or `z\_eff`) |

> ***“Pre-loaded Mode — use this sheet if TFP and efficiency series are already available ... Series will be normalized to $z = \ln(x / x_0) + 1$ if not already done.”***

Running the kit

```
```r
```

```
source("eff_tfp_v3.R")
```

```
Standard mode (full data)
```

```
results <- run_analysis("austria.xlsx")
```

```
Pre-loaded mode (TFP + efficiency already computed)
```

```
results <- run_analysis("portugal.xlsx")
```

```
Built-in Portugal replication (Santos et al. 2021)
```

```
run_portugal_santos2021()
```

### Outputs from Eff-TFP kit

- **Excel workbook** – 9 sheets: elasticities, TFP series, unit root tests, VAR diagnostics, Johansen cointegration (Hz & Hc), intensity constancy, estimated TFP and GDP.
- **PDF report** – publication-ready charts and tables, including the cointegrating vector ( $\beta$ ) and constancy verdict.

These outputs provide the **key empirical relationships** that feed into the IEEM.

---

## 2. Integrated Energy-Economy Model (IEEM, Excel)

### What it does

- Takes the cointegration relationship from the Eff-TFP kit (or uses the pre-calibrated Portugal equations).
- Projects GDP, useful exergy, final exergy, renewable capacity, and CO<sub>2</sub> emissions up to 2050.
- Compares **Business-as-Usual (BAU)** and **Decarbonisation** scenarios (Portuguese RNC2050).
- Models sectoral exergy allocations, carrier-level efficiencies, and renewable investments.

### Relationship to Eff-TFP kit

- The IEEM's TFP equation is directly derived from the cointegration analysis:
  - For Portugal (pre-calibrated):  $A_t = (\varepsilon_t / \varepsilon_{1960})^{1.0480}$

- For any other country, the coefficient ( $\beta$ ) from the Eff-TFP kit (Hz specification, quality-adjusted TFP) should replace the exponent 1.0480 in the BAU and Decarbonization sheets.
- The useful exergy intensity (if found constant) is used to link projected GDP to useful exergy demand. If not constant, the kit provides trend extrapolation options.

### Excel workbook structure

Sheet	Content
Aggregate_Efficiency_BAU	Historical (1960-2022) and projected BAU shares of final exergy by sector, carrier, end-use, and their efficiencies.
BAU	Full BAU scenario: renewable capacities, investment, labour, TFP, GDP, useful/final exergy, CO <sub>2</sub> emissions.
Aggregate_Efficiency_Decarboniz	Same structure as BAU, but input shares follow the Portuguese Carbon Neutrality Roadmap 2050 (RNC2050).
Decarbonization	Full Decarbonisation scenario, including extra renewable capacities and costs relative to BAU.
Charts	Key indicators (efficiency, CO <sub>2</sub> , final energy, GDP, renewable GW, sector shares) for both scenarios.

### How to use IEEM for a new country

1. **Run the Eff-TFP kit** for your country to obtain:
  - Cointegration coefficient  $\beta$  (Hz specification, quality-adjusted TFP).
  - Useful exergy intensity (mean value if constant, or trend).
2. **Open MODEL\_IEEM.xlsx.**
3. **Replace the Portugal-specific TFP equation** in cells BAU!I61:AJ61 and Decarbonization!I68:AJ68 with:

text

$$A_t = (\varepsilon_t / \varepsilon_{\text{base}})^{\beta}$$

where  $\beta$  is the cointegration coefficient from the Eff-TFP kit. The base year is the first year of your historical data.

4. **Replace the useful exergy-GDP link** in cells BAU!I71:AJ71 and Decarbonization!I73:AJ73:
  - If intensity is constant: GDP \* mean\_intensity.
  - If not constant: use the extrapolation formula from the kit.

5. **Update the historical and projected input data** (renewable capacities, demographics, sectoral exergy shares, efficiencies) to match your country.
6. **Run the BAU and Decarbonisation scenarios** – all dependent cells recalculate automatically.

### Outputs from IEEM

- Projected GDP (M€), useful and final exergy (TJ), aggregate exergy efficiency, renewable capacity (GW), CO<sub>2</sub> emissions (Mt), CO<sub>2</sub> intensity.
- Extra investment costs for the Decarbonisation scenario.
- Time-series charts ready for reporting.

### 3. Workflow diagram

text

Historical data (GDP, K, L, X\_final, X\_useful, factor shares)

↓

[Eff-TFP Analysis Kit (R)] →  $\beta$  (cointegration coefficient), useful exergy intensity

↓

[IEEM Excel] → Scenario simulations (GDP, energy, CO<sub>2</sub>)

### 4. Repository contents

File	Description
eff_tfp_v3.R	Main Eff-TFP analysis script
MODEL_IEEM.xlsx	Integrated Energy-Economy Model (Excel)
data_template_v3.xlsx	Template for input data (standard and pre-loaded)
austria.xlsx, usa.xlsx, portugal.xlsx	Example data files
eff_tfp_report_*.pdf	Sample PDF reports
README_EfficiencyTFP_Kit_v3.docx	Detailed Eff-TFP user guide
MAPS_PartB_Final.pdf	MAPS project proposal (background)

### 5. Requirements

#### Eff-TFP kit

- R version 4.1 or later

- Internet connection (automatically installs required packages)

## IEEM

- Microsoft Excel (any version that supports standard formulas, no VBA required)

---

## 6. License

Code and example data © MAPS Project (Horizon Europe, Grant Agreement No. 101137914).

Distributed under the **MIT License** – see LICENSE file for details.

---

## 7. Citation

If you use this module in your research, please cite:

Sousa, T., Santos, J., Felício, L., Gonçalves, J., De Ketelaere, J., Domingos, T. (2026). The Energy and Materials Module. MAPS Deliverable D4.2, Universitat de Barcelona, Barcelona.

Santos, J., Borges, A.S., Domingos, T. (2021). Exploring the links between total factor productivity and energy efficiency: Portugal, 1960–2014. *Energy Economics*, 101, 105407.

De Ketelaere, J., Santos, J., Domingos, T. (2026, under review). Total Factor Productivity is fully explained by changes in physical energy efficiency: an exergy economics approach. *Energy Economics*.

---

## 8. Contact

MAPS Project – <https://mapsresearch.eu/>

Horizon Europe Grant Agreement No. 101137914